

AGILE & SDLC IMPLEMENTATION DOCUMENT

Employee Data Synchronization using SAP Integration Suite

SAP Cloud Integration | SAP SuccessFactors | SAP S/4HANA | SFTP | Email

Methodology: SDLC with Agile Scrum

Document Type: Project Implementation & Delivery Methodology

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1. Introduction

The Employee Data Synchronization using SAP Integration Suite project automates the exchange of employee master data between SAP SuccessFactors and SAP S/4HANA, with supporting SFTP file processing and email-based error notification.

The project follows a structured Software Development Life Cycle (SDLC) combined with Agile Scrum. SDLC provides lifecycle governance, while Agile provides incremental delivery, frequent validation, and continuous improvement.

The solution uses SAP Cloud Integration to retrieve employee information, transform and map the data, deliver it to downstream systems, and provide monitoring and controlled error handling.

2. SDLC Approach

The SDLC framework provides a systematic process for planning, designing, developing, testing, deploying, and maintaining the integration solution.



3. Agile Scrum Implementation

Agile Scrum is used to execute the SDLC activities incrementally. Requirements are maintained in a Product Backlog and delivered through Sprints.

The Agile approach provides continuous feedback, early defect identification, incremental delivery, improved requirement management, and better visibility of project progress.

4. Agile Project Lifecycle



5. Agile Scrum Roles

Product Owner

Defines business requirements, prioritizes the Product Backlog, clarifies acceptance criteria, reviews completed functionality, and provides stakeholder feedback.

Scrum Master

Facilitates Scrum ceremonies, tracks progress, removes blockers, and promotes Agile practices and continuous improvement.

SAP Integration Developer

Designs and develops iFlows, configures adapters and security, implements mappings and transformations, performs testing, troubleshooting, and deployment preparation.

Business Analyst

Translates business requirements into functional requirements, mapping specifications, business rules, and acceptance criteria.

QA / Test Engineer

Prepares test scenarios and data, executes functional and integration testing, validates error handling, documents defects, and performs regression testing.

Technical Architect

Defines integration architecture, security, scalability, maintainability, error-handling strategy, and technical standards.

6. SDLC Phase 1 – Requirement Analysis

The objective of Requirement Analysis is to understand the business need and convert it into clear, testable requirements.

Business Requirement

The business requires an automated mechanism to synchronize employee information from SAP SuccessFactors with SAP S/4HANA. The solution should retrieve employee records, transform them, map required fields, deliver the information, generate files where required, support SFTP processing, notify administrators on failures, and provide monitoring.

7. Non-Functional Requirements

Security: credentials must be maintained securely using SAP Cloud Integration security material and must not be hardcoded.

Reliability: valid records should process successfully and failures should be handled in a controlled manner.

Maintainability: iFlows must be logically organized and understandable to other developers.

Scalability: the architecture should support increasing employee volumes without major redesign.

Monitoring: message processing must be traceable through SAP Cloud Integration monitoring.

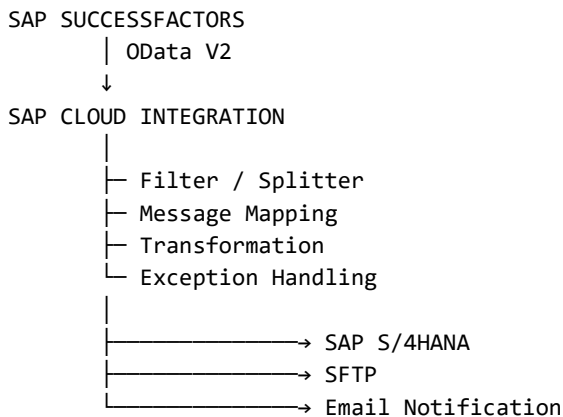
Error Handling: relevant failures must be captured and communicated to responsible administrators.

8. Requirement Traceability

ID	Requirement	Design / Component	Testing
FR-001	Retrieve employee data	SuccessFactors OData	Functional Test
FR-002	Process/filter records	Filter / Splitter	Functional Test
FR-003	Transform data	Message Mapping	Mapping Test
FR-004	Send to S/4HANA	OData Receiver	Integration Test
FR-005	Generate CSV	XML-to-CSV Converter	Output Test
FR-006	SFTP processing	SFTP Adapter	Connectivity Test
FR-007	Error handling	Exception Subprocess	Negative Test
FR-008	Email notification	Mail Adapter	Notification Test

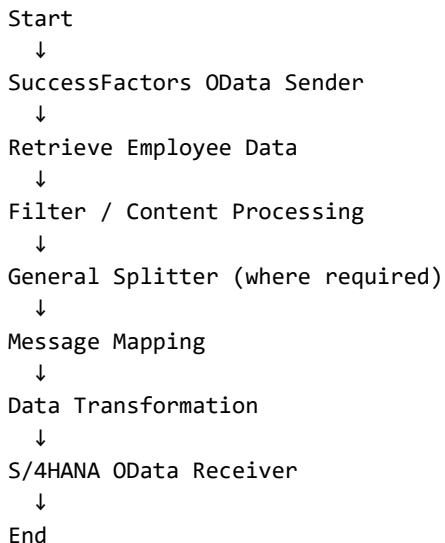
9. SDLC Phase 2 – Solution Design

During the Design phase, requirements are converted into a technical integration architecture.



10. Integration Flow Design

Employee Synchronization iFlow



The flow is designed to separate acquisition, processing, transformation, delivery, and exception handling so that each stage can be tested and maintained independently.

11. Data Mapping Design

Target Field	Source Field	Mapping Logic
Mandt	None	Constant value 500
Participantid	firstName + lastName	Concatenate
Country	country	Direct Mapping
Sfempname	firstName + lastName	Concatenate
City	city	Direct Mapping

Example input: Robert Harrison, US, Dallas. Expected target values: Mandt=500; Participantid=Robert Harrison; Country=US; Sfempname=Robert Harrison; City=Dallas.

12. SDLC Phase 3 – Development

Development converts the approved technical design into SAP Cloud Integration artifacts.

Activities include package and iFlow creation, sender and receiver configuration, SuccessFactors OData setup, authentication, filtering, splitter configuration, source and target structures, Message Mapping, transformation, S/4HANA connectivity, SFTP, Mail Adapter, Exception Subprocess, unit testing, defect fixing, and deployment preparation.

13. Agile Sprint Plan

Sprint	Focus	Key Deliverables
Sprint 1	Requirements & Architecture	Requirements, user stories, architecture, mapping specification
Sprint 2	SuccessFactors Integration	OData configuration and source data validation
Sprint 3	Mapping & Transformation	XSDs, mapping, filtering, splitter, target validation
Sprint 4	S/4HANA Integration	Receiver configuration and end-to-end synchronization
Sprint 5	SFTP & Error Handling	SFTP, exception subprocess, email notification
Sprint 6	Testing & Deployment	Test evidence, defect closure, documentation, release

14. Product Backlog & User Stories

Story ID	User Story	Priority	Status
US-001	Retrieve employee data from SuccessFactors	High	Done
US-002	Filter required employee records	High	Done
US-003	Transform employee data	High	Done
US-004	Map employee fields	High	Done
US-005	Send data to S/4HANA	High	Done
US-006	Generate CSV output	Medium	Done
US-007	Transfer files through SFTP	Medium	Done
US-008	Implement exception handling	High	Done
US-009	Send error notification	High	Done
US-010	Perform integration testing	High	Done
US-011	Prepare documentation	Medium	Done
US-012	Deploy integration flow	High	Done

Sample User Story – US-001

As an integration administrator, I want the platform to retrieve employee information from SAP SuccessFactors so that employee master data can be synchronized with downstream systems.

Acceptance Criteria: connectivity is successful; employee records are retrieved; required fields are available; invalid authentication produces a controlled error; retrieved data reaches the next processing step.

15. Definition of Ready / Done

Definition of Ready

A story is Ready when the requirement is understood, source and target systems are identified, fields and mapping rules are available, acceptance criteria are documented, dependencies are known, and test data is available.

Definition of Done

A story is Done when development, configuration review, unit testing, functional testing, defect resolution, acceptance criteria, error handling, monitoring, documentation, and deployment readiness are completed.

16. Agile Ceremonies

Sprint Planning

The team reviews the Product Backlog, selects stories, defines the Sprint Goal, estimates effort, identifies dependencies, and creates the Sprint Backlog.

Daily Scrum

Team members communicate completed work, planned work, and blockers.

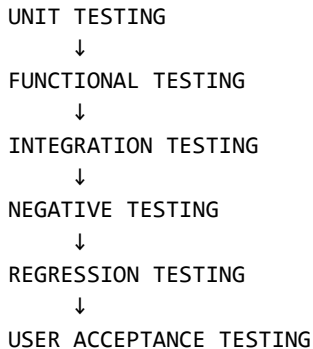
Sprint Review

Completed functionality is demonstrated to stakeholders and feedback is captured.

Sprint Retrospective

The team identifies what went well, what did not go well, and what should be improved in the next Sprint.

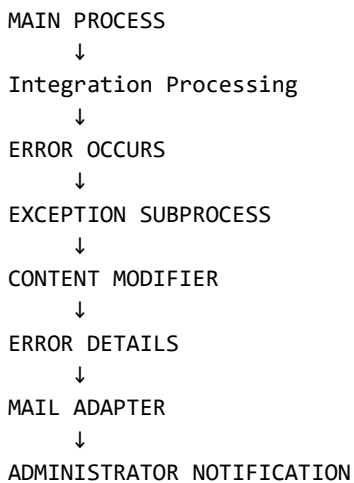
17. Testing Strategy



Unit testing validates individual components such as adapters, filters, splitters, mappings, converters, SFTP, and Mail Adapter. Functional testing validates business rules. Integration testing validates end-to-end communication. Negative testing validates controlled failure behavior.

18. Error Handling

The solution uses an Exception Subprocess to capture runtime failures and initiate controlled error processing.



Notifications can include the iFlow name, timestamp, message ID, processing status, error message, and relevant business information.

19. Defect Management

Defect ID	Description	Severity	Status
DEF-001	Target Participantid not generated	High	Resolved
DEF-002	Target XSD occurrence mismatch	High	Resolved
DEF-003	SFTP host key rejected	High	Resolved

DEF-004	Missing employee field	Medium	Resolved
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Root-cause analysis should be performed before fixing defects. For mapping errors, verify source XSD, target XSD, occurrence settings, mapping context, mandatory fields, and generated target XML.

20. SDLC Phase 5 – Deployment

After successful testing and approval, the integration solution is deployed.

Deployment activities: final configuration review, security validation, endpoint verification, parameter verification, test evidence review, approval, deployment, activation, smoke testing, and post-deployment monitoring.

Deployment Checklist

- ☐ Package and iFlow completed
- ☐ Credentials configured securely
- ☐ Endpoints validated
- ☐ Message mapping validated
- ☐ Exception handling tested
- ☐ SFTP configuration validated
- ☐ Email notification tested
- ☐ Functional and integration testing completed
- ☐ Documentation completed
- ☐ Approval received
- ☐ iFlow deployed and activated
- ☐ Smoke test completed

21. SDLC Phase 6 – Maintenance

After deployment, maintenance includes monitoring, troubleshooting, mapping updates, credential or endpoint changes, support for new fields, performance optimization, security updates, defect resolution, and enhancement requests.

22. Production Monitoring

SAP Cloud Integration monitoring is used to monitor message processing and identify failures.

Important information includes message status, processing duration, message ID, error details, sender, receiver, timestamp, and processed record count.

Typical processing states include COMPLETED, FAILED, RETRY, and PROCESSING.

23. Agile Metrics

Sprint Velocity

Velocity measures the amount of work completed during a Sprint. Example: Sprint 1 = 20 story points; Sprint 2 = 25; Sprint 3 = 30.

Burndown

A Sprint Burndown tracks remaining work over Sprint days. The desired trend is a steady reduction toward zero remaining work.

24. Risk Management

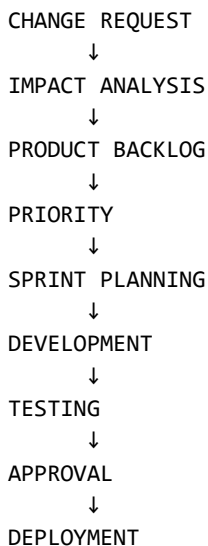
Risk	Impact	Mitigation
Source system unavailable	High	Test connectivity early
Incorrect credentials	High	Validate security material
Mapping mismatch	High	Validate source/target XSDs
Missing mandatory data	Medium	Validation and error handling
SFTP connection failure	High	Validate host key and credentials
Target service unavailable	High	Monitoring and retry strategy
Requirement changes	Medium	Manage through Product Backlog
Deployment issue	High	Deployment checklist and smoke test
Production failure	High	Monitoring and alerting

25. Security Approach

Credentials should be stored in SAP Cloud Integration Security Material and never hardcoded in integration logic.

Communication should use secure protocols such as HTTPS and SFTP with appropriate authentication. Access to development, deployment, and monitoring functions should follow least-privilege principles.

26. Change Management



This process ensures that requirement changes are controlled and traceable instead of being introduced without impact analysis.

27. Documentation Strategy

Project documentation should include the Business Requirement Document, Functional Specification, Technical Design, Mapping Specification, source and target XSDs, Test Plan, Test Results, Defect Report, Deployment Checklist, Operations Guide, Troubleshooting Guide, Agile Summary, and Final Project Report.

28. Acceptance Criteria

The project is accepted when SuccessFactors connectivity and employee retrieval are successful; filtering and mapping produce expected data; S/4HANA receives the required information; CSV/SFTP processing works where applicable; exception handling and email notification work; testing is completed; critical defects are resolved; documentation is complete; and deployment monitoring is verified.

29. Continuous Improvement

Continuous improvement is driven by production monitoring, user feedback, defect analysis, performance observations, new business requirements, security requirements, and SAP Integration Suite enhancements.

Potential future enhancements include additional employee attributes, more SuccessFactors entities, automated retries, advanced monitoring dashboards, centralized logging, reconciliation, enhanced alerting, and performance optimization.

30. Conclusion & Appendices

The Employee Data Synchronization project combines SDLC governance with Agile Scrum execution to provide a structured, iterative, and enterprise-oriented integration delivery model.

SDLC provides the lifecycle structure: Requirement Analysis → Design → Development → Testing → Deployment → Maintenance. Agile provides iterative execution: Product Backlog → Sprint Planning → Development → Testing → Review → Retrospective → Continuous Improvement.

The approach results in a secure, testable, maintainable, monitorable, documented, and production-ready SAP integration solution.